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TAPS –A 4th Generation Propulsion Combustor Technology for Low Emissions

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ABSTRACT

This paper describes a low-emissions combustion system that uses a Twin Annular Premixing Swirler (TAPS) arrangement to further reduce NOx emissions without adversely affecting other conflicting design requirements of propulsion engine combustors. Two combustion systems have been developed using the test rigs: a single annular combustor scaled for the CFM56 engine size, Tech56/CFM SAC TAPS; and a Dual Annular TAPS for the GE90 size engine, DAC TAPS. Both the DAC and SAC TAPS give approximately 50% reduction in the landing/takeoff (LTO) NOx compared to current rich-dome product combustors. The Tech56/CFM TAPS combustion system has successfully completed engine demonstration including emissions, performance, operability and endurance.

INTRODUCTION

The turbopropulsion combustion design and technology community effort started in the middle and late sixties for the reduction of gaseous and smoke emissions. The first major NASA sponsored initiatives, Experimental Clean Combustor Program, ECCP and Pollution Reduction Technology Program, PRTP[1-4] have been followed by several industry and government funded technology programs including the most recent High Speed Civil Transport (HSCT), Advanced Subsonic Transport (AST) and Ultra Energy Efficient Technology (UEET) initiatives of NASA.

The Sixth Meeting of the Committee on Aircraft Noise May 11, 1981 recommended publication of Environmental Protection Annex 16 Volume II Aircraft Engines. The Second Meeting of the Committee on Aircraft Engine Emissions, June 30, 1981 introduced Volume II relating to control of smoke and gaseous emissions which was adopted by the First Meeting of the Committee on Aviation Environment Protection (CAEP), March 4, 1988, denoted as CAEP1. The details are provided in Reference 5. NOx stringency has subsequently been adopted by CAEP2 and CAEP4 in 1993 and 1999, respectively. For example the following three equations for the Landing and Takeoff NOx, g/kN over the LTO cycle:

LTO NOx = 40 + 2.0 OPR - CAP1 (1)
LTO NOx = 32 +1.6 OPR - CAEP2 (2)

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For engines w/ OPR ≥ 30 and >89 KN rated takeoff thrust
LTO NOx = 7 + 2.0 OPR - CAEP4 (3)

Here OPR denotes the engine overall pressure ratio. For example, for 30 OPR, the above three equations give LTO NOx values of 100, 80 and 67, respectively. In other words, CAEP 2 reduced NOx by 20% over CAEP1, and CAEP4 reduced it by another 16%.

Recent document of ICAO/CAEP/Working Group 3 (WG3-WP3/16,ICAO/CAEP/WG3, 14-15 May 2002, Dallas, Texas) states:
CO_2 and NOx emissions are important for global climate change. Any future ICAO/CAEP action (and technology goals) must consider fuel efficiency (CO_2 reduction) and simultaneous decrease of NOx emissions. Reducing NOx is important because of the climatic effects of the ozone produced at flight altitudes.

An atmospheric science paper presented by the UK at the recent ICAO/CAEP Working Group 3 meetings indicates that NOx from aircraft is much more important to global warming than CO_2. Previous assessments have suggested that for aircraft, the effect of NOx is potentially equal to that of CO_2.

Highlights of a recent GAO study, GAO-03-252, presented to the Chairman, Subcommittee on Aviation, the US Congress House Committee on Transportation and Infrastructure lists:
"Aviation related emissions of nitrogen oxides, which contribute to formation of ozone, have been of particular concern to many airport operators. A federal study at 19 airports estimated that by 2010, aircraft emissions have the potential to significantly contribute to air pollution in the areas around these airports. ------- GAO recommends that the Federal Aviation Administration (FAA) develop a strategic framework that addresses the need for information on the extent and impact of emissions, identifies reduction options, establishes goals and time frames for achieving needed reductions, and defines the roles of government and industry in developing and implementing reduction programs."

NOx REDUCTION TRENDS

Propulsion engine industry has been putting considerable effort for reducing fuel consumption and NOx emissions index, NOxEI. Both of these impact LTO NOx. As shown in Fig. 1a, LTO NOx has come down by approximately 45%, whereas takeoff NOxEI, which is

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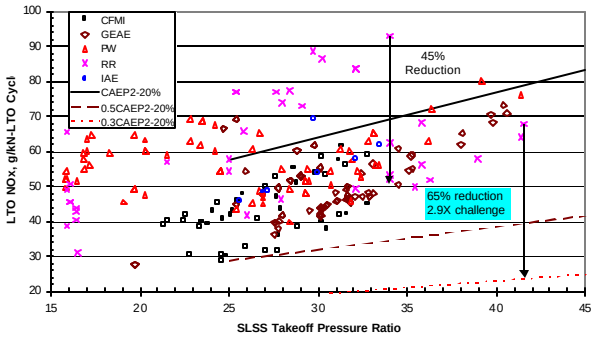
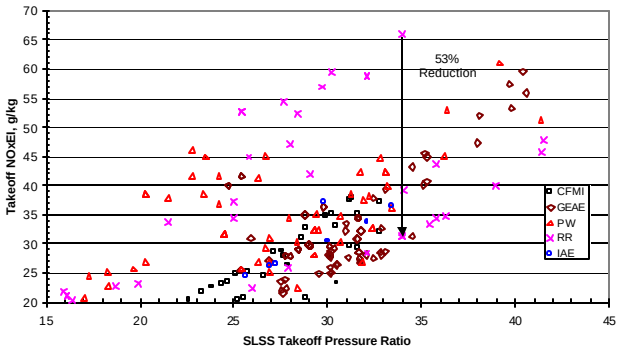
indicative of combustors' NOx technology level is down by approximately 53%, as presented in Fig. 1b; or LTO NOx divided by climbout specific fuel consumption, Figure 4.



(a)



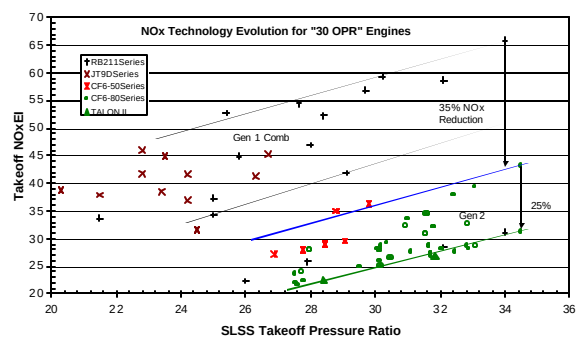
(b)

Figure 1: Propulsion engine NOx emissions levels.

From Figures 1a and 1b, one can see that approximately 50% NOx emissions reduction achieved in propulsion engines may not appear as impressive as 90%+ reduction achieved in natural gas-fired industrial aero-derivative engines with water injection or Dry-Low Emissions (DLE) technology[7]. This difference can be explained by the fact that propulsion engine combustion systems pose special design challenges in regard to weight, life (which is driven mostly by low-cycle fatigue considerations), compactness, reliability, wider range of operating envelope, engine acceleration, ground and altitude lean blowout and starting requirements.

However, NASA's HSCT, AST and UEET initiatives have challenged the industry to propose and develop innovative propulsion engine combustion technologies to achieve 50% and 30% of CAEP2 NOx levels without adversely impacting other performance, emissions, operability and durability requirements. It is a common industry practice to have approximately 20% margin for engine combustion products to allow for option to go with single engine or three engine emissions measurement, scale-up uncertainty and thrust growth or de-rate for subsequent engine models. As shown in Fig. 1a, only 8 engine models are close to meeting 50%CAEP2 NOx technology goal within approximately 20% margin.

On the other hand in order to meet 30%CAEP2 levels, we need to get additional 65% NOx reduction on the top of 45% reduction already achieved, refer to Figure 1a. Considering it took the industry approximately 30 years to achieve 45% reduction in NOx, one wonders how long it will take to achieve another 65% reduction especially with no obvious financial incentives for reducing NOx emissions. Figure 2a further elaborates the challenge of achieving NOx reduction in propulsion engine by focusing on 5 engine models, namely RB211, JT9D (belonging to Generation 1 low-emissions combustors), CF6-50, CF6-80, TALON II, and RB211 from Gen 2 low-emissions technology. As shown, 35% NOxEI reduction is achieved going from Gen 1 to Gen 2 technology, and that all the four Gen 2 combustors fall within a band of 25%. Fig. 2a shows the NOx technology evolution of nominal 30 overall pressure ratio (OPR) engines over a period of 30 years.



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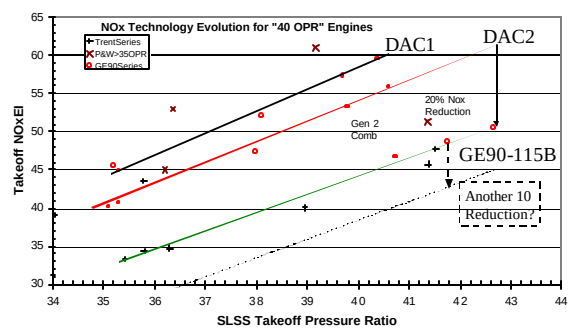


(b)

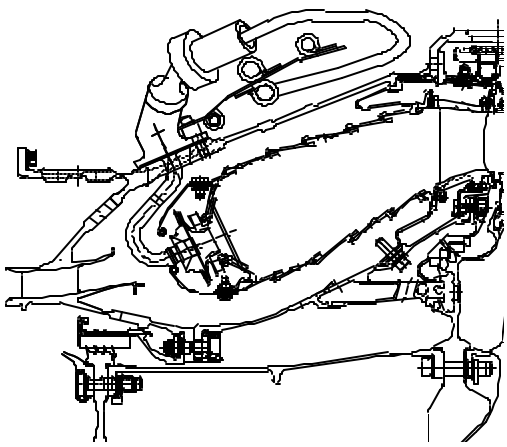
Figure 2: NOx emissions technology evolution of selected engines.

The development of turbopropulsion engines of nominal 40 OPR shown in Fig. 2b started in the early 90's, and clearly benefited from the emissions technology base of 30 OPR engines. The lowest NOxEI

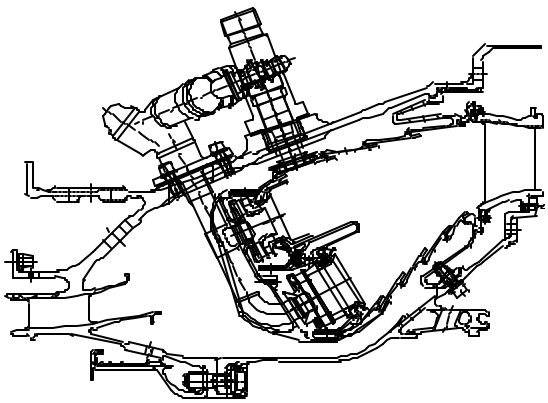
data points of 30 OPR including TALON II, Trent 800 and GE90-115B fall along a narrow band and form the basis of the best rich-dome low-emissions combustion (LEC) technology called Gen 2. The low-emissions combustion design community is trying to get additional 10% reduction from this level by carrying out further refinements available within the design space.

DAC – GEN 3 TECHNOLOGY

The first successful adaptation of lean-dome design to product is the Dual Annular Combustor[1] (DAC) conceived in early 1970's to the CFM56-5B, 5B/P and 7B engine models introduced during 1995-1998 timeframe. As shown in Figure 3, compared to Single Annular Combustor (SAC), DAC has two combustion domes, the outer dome optimized for operability and low-power operation, and the inner for high-power NOx.



(a) Single annular combustor schematic.



(a) Dual annular combustor schematic.

Figure 3: Schematics of the CFM56-5/7 single and dual annular combustors.

Introduction of DAC into product, creating Generation 3 low-emissions combustors family, met the design intention, i.e. of further reducing NOx, as shown in Figure 4. Here data of landing-takeoff NOx emissions normalized with climb-out specific fuel consumption is presented as a function of takeoff pressure ratio. This process reduces the data scatter of different engines, which invariably happens when plotted in terms LTO NOx values. For example, LTO NOx data of the CFM56-2/3, CF6-80C2/E LEC, Trent 800 and TALON II falls along one curve identified as LEC (Gen 2) with mean square root error, σ of approximately 1.4%. The GE90-115B NOx data also falls on the same curve. Similarly, LTO NOx data of the lean DAC's used in the CFM56-5, 5B/P and 7B engine models form a curve denoted as Lean DAC (Gen 3) with σ of 2.3%. There is however more scatter in the PreLEC (Gen 1) engine data because the technology base of these engines evolved over a number of years and no attempt was made to compile data from engines with comparable technology levels.

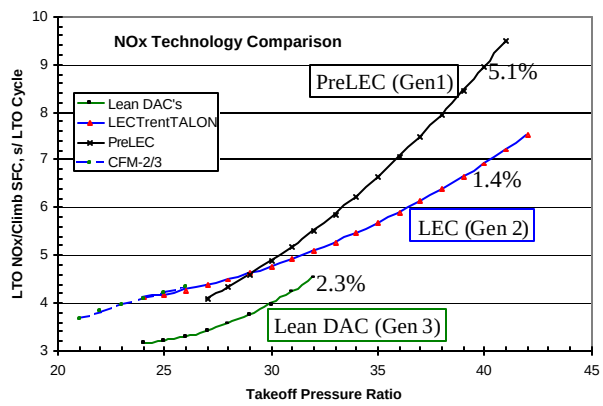


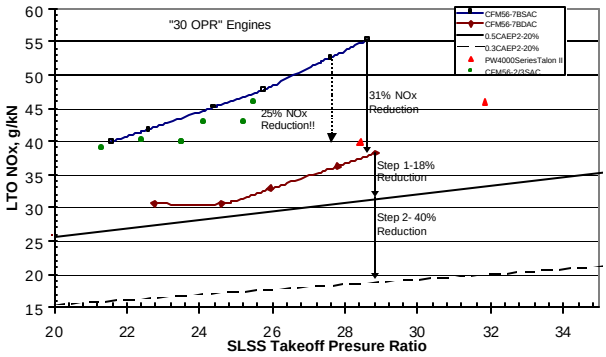
Figure 4: Generations 1, 2 and 3 NOx technologies compared in terms of LTO NOx/climbout SFC.

TAPS Combustion System

Considering the significant level of emissions reduction of the past 30 years has been based on optimizing fuel-rich mixer, e.g., swirl cup, and extending the same basic mixer (rich or lean) for DAC application, it was important to recognize where we have been and what should we set as our target for next-generation low-emissions combustion technology concepts. NASA AST and UEET program set the realistic goals, i.e., 50% and 70% reduction from CAEP2 levels, Equation 2. As mentioned before, it is a common industry practice to maintain 20% margin from the technology goal for product introduction This means for industry to claim 50% CAEP2 capability at 30 OPR, it is not enough to demonstrate LTO NOx value of 40 g/kN, it should have another 20 margin on the top of it, i.e., LTO NOx value of 32 g/kN. Similarly for 40 OPR, the corresponding values are 48 and 38.4, respectively. On the other hands to achieve 30% CAEP2 technology capability for

product, one needs to demonstrate LTO NOx levels of 19.2 and 23, respectively for 30 and 40 OPR engines. Achieving these levels of NOx emissions might not be possible by adopting evolutionary improvements over the design concepts pursued during the last 30 years including rich or lean swirl cup-based SAC or DAC combustors.

Fig. 5a shows evolution of rich SAC NOx technology for a 30 OPR engine, e.g., the CFM56 engine series. Comparing the CFM56-7B and CFM56-2/3 engine families, the latter was certified in 1983, it has been speculated that further reduction in rich-dome SAC NOx (of approximately 25%) should be possible. Since having achieved approximately 30% reduction with lean CFM56-7B DAC after several years of combustion product development, further development of DAC combustion concept was considered to have limited long-term potential, especially in regard to achieving 30% CAEP2-20% margin. In addition, the lean DAC development experience for "40 OPR" engine, i.e. GE90 DAC1) shown in Fig. 2b, clearly showed that the cooling air requirement even with multi-hole liner cooling technology was too high resulting in the primary zone equivalence ratio close to 0.9 at the takeoff operating conditions. It is for this reason that lean DAC1 combustor has higher NOx than the rich-dome DAC2, Fig. 2b. Other design challenges posed by lean DAC designs are profile, pattern factor, cost, weight, durability and low-power CO and HC emissions.



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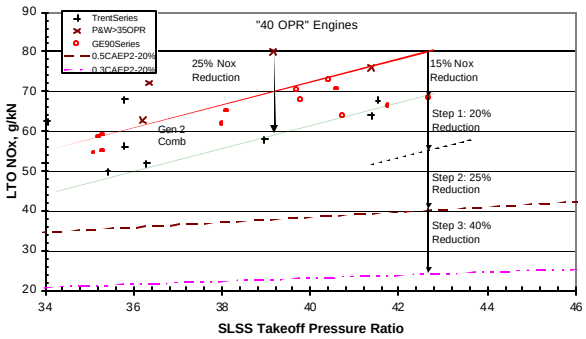

(b)

Figure 5: NOx reduction technology steps to achieve 50% and 30% CAEP2 objectives.

Generation 2 (LEC) combustion technology has been achieved after considerable effort ($ and manpower) expended during the last 30 years. The most recent effort of the last 15 years on 40 OPR engine family (Fig. 5b) shows that we have achieved approximately 25% reduction in NOx. The lowest LEC technology comprised of Trent 800 and GE90-115B is approximately 15% lower than GE90 DAC1 combustor. The overall reduction of 25% from the maximum to LEC technology level has been achieved after considerable design and technology optimization.

In summary, one can conclude based on historical data that typical jumps in NOx reduction technology have been between 10-20% and several years have taken in developing and incorporating new technologies into product. For 30 OPR engines, one needs to develop technology in two steps (stages) with expected NOx reduction levels of approximately 18% and 40%, respectively (from the best available combustion technology, BACT) to achieve emissions levels capability of 50% CAEP2 and 30% CAEP2. On the other hand, for 40 OPR engines, one needs to develop the NOx technology in three steps (Fig. 5b) requiring approximately 20%, 25% and 40% reduction in NOx emissions starting from BACT.

In view of these requirements and the lessons learned from the low-emissions propulsion engine combustion technology and product introduction effort of last 30 years (including References 1-4 and 7), the design and development of a new concept (Mod 2 TAPS), Fig. 6 was initiated for SAC (under GE sponsorship) and DAC sponsored under NASA Advanced Subsonic Transport (AST) initiative in March 1998.

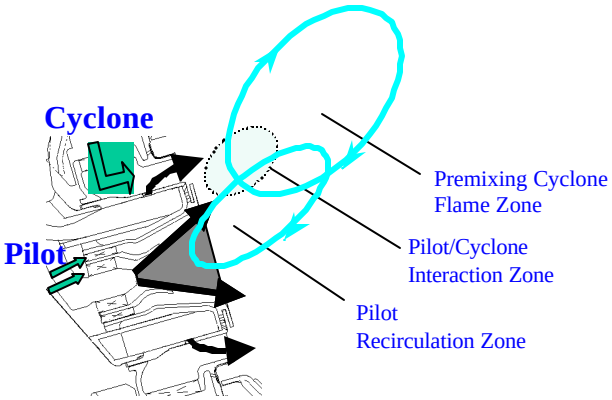

Figure 6: Twin Annular Premixing Swirler (TAPS) Mod 2 schematic.

TAPS Mod 2 produces two co-annular swirling jets produced respectively by pilot and main mixer. Each of the two swirling streams could be the result of multi-swirler arrangement designed specifically for achieving specific design objectives. For example, the pilot uses a high flow number simplex pressure atomizer surrounded by two co-rotating swirlers that help produces spray quality suitable for starting and low-power operation along with the flow field characteristics appropriate for meeting the design requirements for ignition, starting, lean flame stability, combustion efficiency etc. The degree of mixedness of the flowfield influenced mostly by the pilot design features needs to be matched properly with these requirements in addition to providing NOx low enough to meet 50% and 30% CAEP2 requirement. The design features of the main mixer and the pilot have evolved over the years as shown typically in Fig. 7; Fig. 6 shows one of these designs where the main mixer consists of a cyclone or radial inflow swirler, a cavity, and fuel injected by discrete transverse jets. More discussion follows under the section entitled Future TAPS Activities.

The pilot swirling jet(s) controlled initially by the fuel nozzle geometry interacts with the main swirling jet(s) separated suitably by a step height to provide thermal environment required to meet combustor's critical design requirements. All the combustor air except cooling air required for the dome and liners passes through the pilot and cyclone swirlers. Fuel staging between the pilot and main is done within the fuel nozzle by means of "controlled pressure fuel nozzle" logic which allows predetermined discrete flow splits that vary from 100% through the pilot at low-power to 5-10% at the maximum power.

TAPS combustion system aero design and development process relied extensively on systematic use of rig (single module, sector and full-annular test rigs) and engine testing. The state-of-the-art CCD-based design tools[8] provided insignificant pre-test insight during the development phase of TAPS; however, they were useful for providing qualitative understanding of the test results.

Combustion system testing included documenting data for gaseous emissions, smoke, combustion efficiency, pressure drop, profile, pattern factor, dome and liner temperature levels at critically selected points on the dome and liners, ignition, flame propagation at simulated sea-level to altitude operation, and lean blow-out at selected design points.

The Tech56/CFM TAPS SAC combustion system (Fig. 8a) was designed to fit within the spatial envelope of the CFM56-7B engine combustion system (Figure 3a); like wise for the GE90 DAC TAPS (Fig. 8b) consistent with that of the GE90-94B envelope, Fig. 8c. The photographs of the combustors and fuel nozzles for the SAC and DAC TAPS are shown in Figures 9a and 9b. The CFM TAPS testing included engine performance, emissions, inclemental weather and endurance (4000 modified blade cycles) testing on a development CFM56-7B engine, Figures 10a and 10b. The Tech56/CFM TAPS SAC testing effort has accumulated a total of 840 engine hours. The testing for the DAC TAPS has been conducted in a full-scale annular combustor rig with a rig development fuel nozzle. A summary of the test results is given in the following paragraphs.

TAPS TEST RESULTS

Extensive rig testing of both SAC and DAC TAPS has been conducted since 1999 in addition to engine testing of the SAC TAPS combustion system including the development control system. These results have been very encouraging in meeting expectations in regard to pressure drop, combustion efficiency, dome/liner wall temperature levels and gradients, combustor exit temperature quality including profile and pattern factor, lean blowout, ignition and flame propagation at critical design points in FAR simulated environment limited by test facility capabilities. Thorough testing of the Tech56/CFM SAC TAPS development engine testing have identified areas for further development including the occurrence of benign cracks on the dome, fuel nozzle shrouds and locally behind the igniter; all these are considered fixable as part of SAC TAPS product development effort. Engine ground starts (ignition, flame propagation, engine response to throttle push and chop) limited to minimum engine inlet temperature of approximately 15F, and the response in the hailstorm have been considered acceptable. However, the SAC TAPS engine has not gone through any flying test bed demonstration for operability including windmill start, and starting under extreme cold air and fuel conditions.

Combustor exit temperature quality, exhaust smoke, liner wall temperature levels and gradients are generally superior to that of rich-dome SAC/DAC and lean DAC combustors. The gaseous emissions results are presented in Figures 11 and 13.

Since the SAC TAPS emissions data is from a development engine using the ICAO recommended procedure for measuring engine emissions, a back to back to comparison with the CFM56-7B rich SAC and lean DAC is straight forward as shown in Fig. 11. For the CFM56-7B 27000 lb thrust rating, the SAC TAPS system gave 46% and 22% lower NOx compared to its counterpart rich-dome SAC and lean-dome DAC, respectively. HC emissions are essentially comparable

with rich-dome SAC, although slightly higher values trend at the lower takeoff pressure ratio appears evident. CO emissions though lower than DAC is clearly higher than that of rich-dome SAC. There is no plan to further reduce LTO CO emissions for 30 OPR engines other than what needs to be required for certification of low-thrust engine models, since most of the future high energy efficient engines may not operate below 25 OPR. Instead the emphasis may be more on reducing the operating T3 window where combustion efficiency at the intermediate power could fall below 99.5% due to fuel staging at intermediate power levels, as shown typically in Figure 12. The pilot-fueled only mode can be extended beyond 700F if one does not mind NOx emissions during this operation be comparable with rich-dome SAC, as long as fuel nozzle circuits film temperature levels are below the required critical levels to avoid fuel coking and deposition. Nevertheless, intermediate power mode fuel staging area needs to be explored further to achieve optimum trade between fuel nozzle coking/plugging/deposition, NOx, combustion efficiency, etc.

Extensive full-scale rig database of the GE90 DAC TAPS was used to develop emissions correlations as a function of P^n, Exp(T3/b) and FAR^c for each of the fuel staging modes. The data was taken for simulated engine operating conditions with T3, P3 and fuel/air ratio (FAR) as high as 1170F, 350 psi and 0.037, respectively. The rig was however run for the engine operating conditions (P3, T3 and Wa3) required to get data for 7% and 30% power points which are the ICAO idle and approach conditions. These data along with correlations were used to calculate LTO NOx, HC and CO emissions of the AST DAC TAPS cycle as well as GE90-94B DAC TAPS, which are compared with engine data shown in Fig. 13. The GE90-94B DAC TAPS gives 54% lower NOx compared to the rich-dome DAC, the current product combustor and simultaneously CO and HC emissions are comparable with the product engines.

FUTURE TAPS ACTIVITIES

As summarized in the last section, the original design intent (nested flame whose strength can be varied throughout the operating envelope), the underlying functionalities, and feasibility of TAPS as a viable low-NOx emissions combustion system concept has been clearly demonstrated for both SAC and DAC application. It has shown that 50% NOx reduction over current product combustors can be achieved without adversely impacting other design requirements including performance, operability and durability. There are still other challenges ahead, e.g., operability, fuel nozzle purging without causing plugging/deposition, product cost and weight for finding optimum engineering solution as part of product development; however, for technology the focus should be how much further NOx reduction can be achieved by refining the basic TAPS concept. Some of the refinements identified in Fig. 7 have the potential to reduce TAPS NOx by additional 50% (2X challenge) and 75% (4X challenge). These advances are necessary if we were to achieve step 2 (identified in Fig. 5a), and steps 2 and 3 of Fig. 5b, if we ever want the combustion products for 30 and 40 OPR engines to have 30% CAEP2 NOx capability with 20% margin.

There are several technology enablers that need to be identified and developed during the next 10 years for potential enhancements to the 1st generation TAPS combustion systems tested to-date for potential application in Tech56/CFM and GE90 engine families. A relative comparison of 2X and 4X reduction in NOx emissions index for 30 and 40 OPR engines over the current SAC and DAC TAPS and LEC is given in Figures 13a and 13b for the four ICAO LTO points of generic 30 and 40 OPR engine families.

SUMMARY

A brief overview of the low-NOx emissions technology and product evolution of the last 30 years was given leading to the rationale for starting with formulation of a new strategy for developing an affordable ultra-low emissions propulsion engine combustion system based on Twin Annular Premixing Swirler (TAPS) with a near-term intention being “Lean DAC Emissions with SAC Simplicity”.

The main objective of the 1st phase TAPS effort for potential application in Tech56/CFM and GE90 class engines of SAC and DAC TAPS has been successfully concluded; the former through an extensive engine demonstration, the latter only through a full-scale annular combustor tested under simulated engine operating conditions.

Approximately 50% reduction in LTO NOx over the current product combustors was achieved without adversely impacting other design requirements including performance, operability and durability etc. Further near-term TAPS technology development effort is planned to continue to make it more attractive for new centerline engine combustors; in addition to reducing NOx by additional 50% and 75% in order to quantify the TAPS entitlement levels.

ACKNOWLEDGEMENT

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- Approximately 200 individuals for their contribution in their field of expertise including Combustion Aero, Mechanical and Thermal Design; Fuel System, Controls, Case, and Configurations Design; Materials and Special Manufacturing Processes; Hardware Procurement, Instrumentation, Assembly and Test; and Engine System Design Integration.
- The TAPS CCD Tools team.

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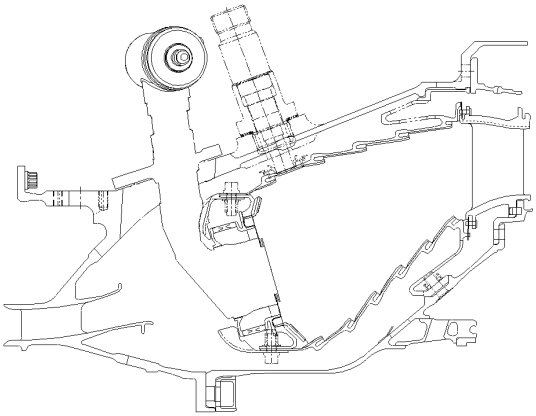
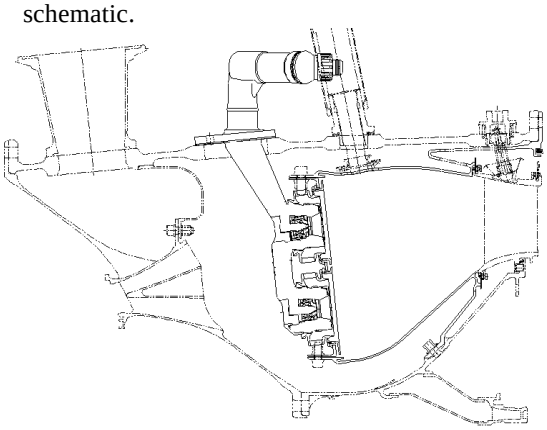
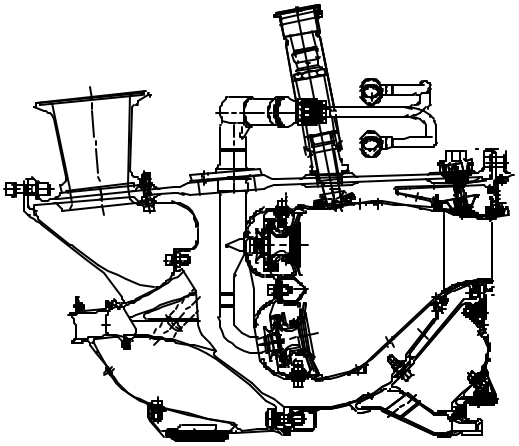


(a) Tech56/CFM TAPS combustion system schematic.

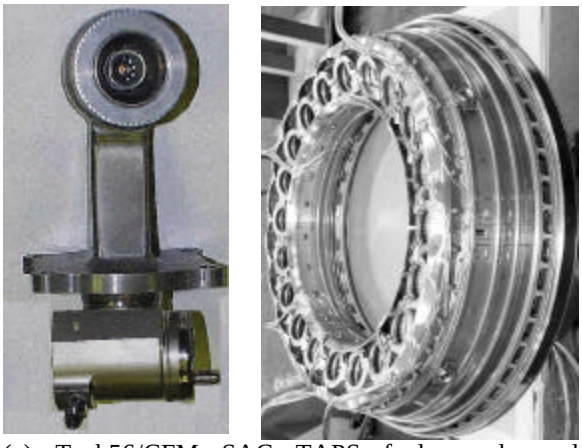


(b) GE90 DAC TAPS combustion system schematic.

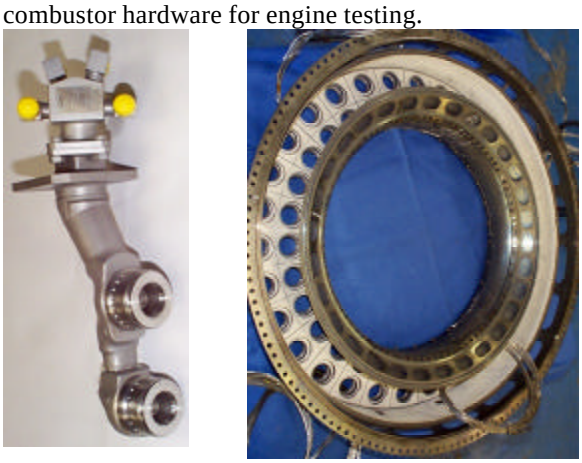


(c) Rich dome GE90-94B DAC combustion system schematic.

Figure 8:Tech56/CFM SAC and GE90 DAC TAPS combustion system schematics.



(a) Tech56/CFM SAC TAPS fuel nozzle and combustor hardware for engine testing.



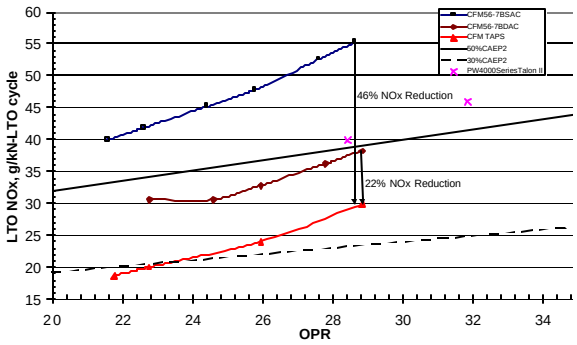
(b) GE90 DAC TAPS combustor and fuel nozzle tested in a full-scale annular combustor rig to 350 psi, 1170F inlet temperature and 0.037 fuel/air ratios.

Figure 9: SAC and DAC TAPS combustors and fuel nozzles hardware tested.

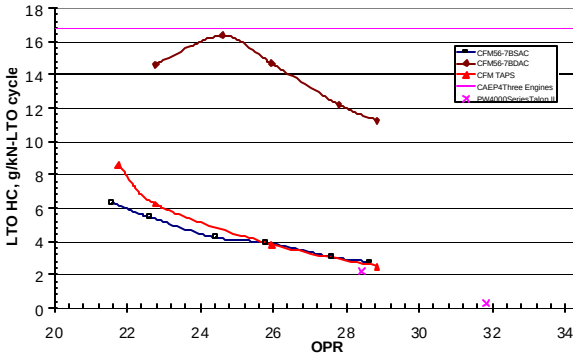
(a) CFM56-7B development engine with SAC TAPS combustion system.

Tech56/CFM56 SAC TAPS engine running in Peebles hailstorm test facility.

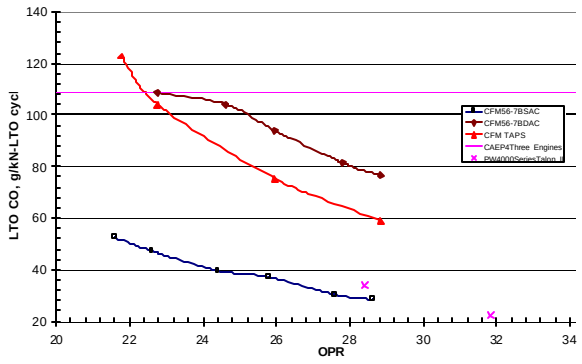
Figure 10: Emissions, performance, ground start, accel/decel and inclemental weather testing of the Tech56/CFM56 SAC TAPS in a development engine.



(a) Landing Takeoff (LTO) NOx.



(b) LTO hydrocarbons



(c) LTO CO

Figure 11: Tech56/CFM TAPS engine emissions compared with currently certified products.

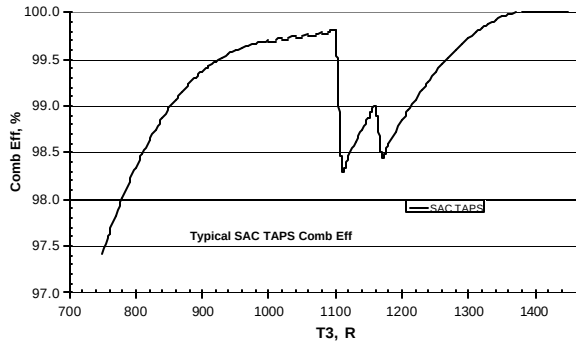
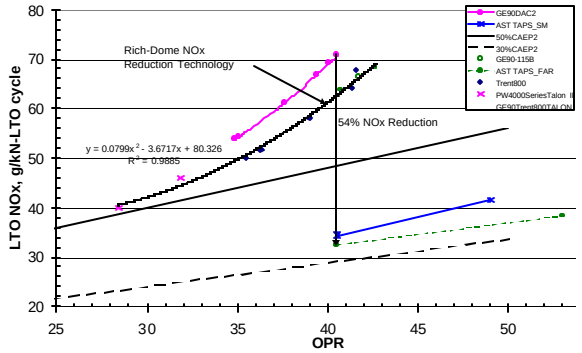
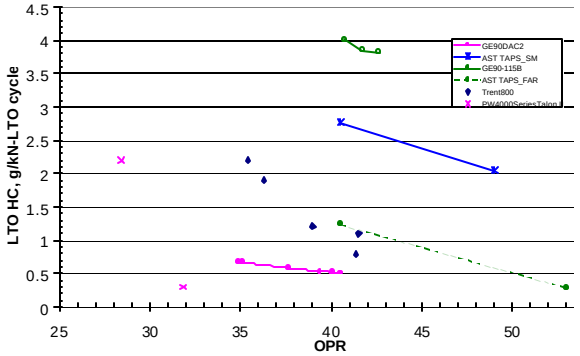


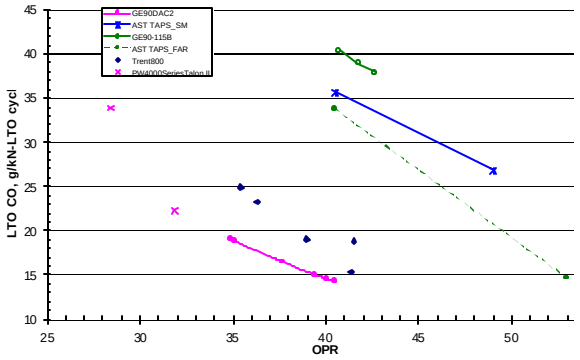
Figure 12: Typical Tech56/CFM TAPS system combustion efficiency.



(a) Landing-Takeoff NOx

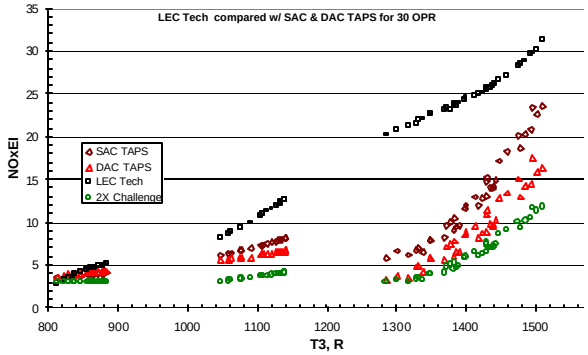


(b) LTO hydrocarbons

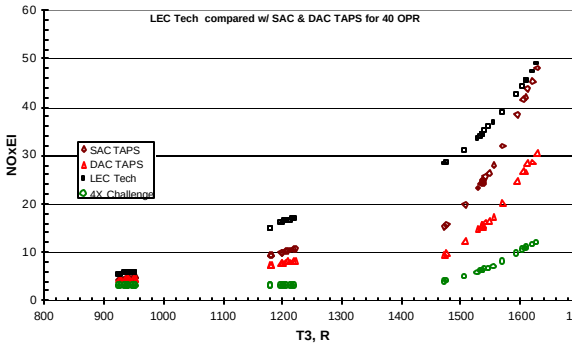


(c) LTO CO

Figure 13: AST (GE90) DAC TAPS LTO emissions calculated from full-annular rig data



(a) 50% further NOx reduction (2X challenge) for 30 OPR engines.



(b) 75% further NOx reduction (4X) from CFM TAPS for 40 OPR engines.

Figure 14: Potential 2X and 4X NOx reduction with basic TAPS SAC concept.

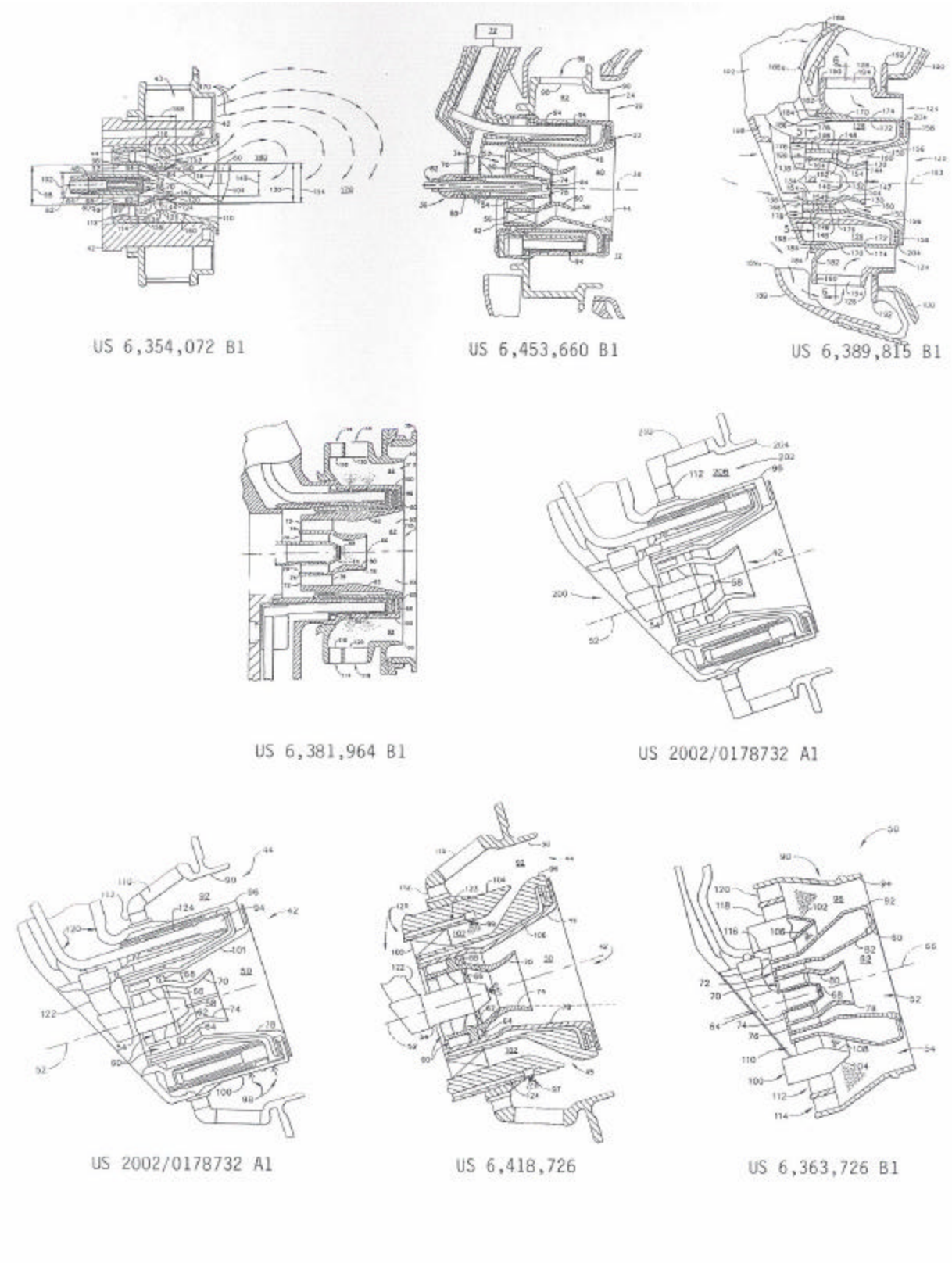


Figure 7: TAPS mixer designs issued as US Patents.